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RESEARCH ARTICLE | *Physical activity and inactivity*

Aerobic system analysis based on oxygen uptake and hip acceleration during random over-ground walking activities

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Beltrame T, Hughson RL. Aerobic system analysis based on oxygen uptake and hip acceleration during random over-ground walking activities. *Am J Physiol Regul Integr Comp Physiol* 312: R93–R100, 2017. First published November 16, 2016; doi:10.1152/ajpregu.00381.2016.—Deteriorated aerobic response to moderate exercise might precede the manifestation of clinical symptoms of noncommunicable diseases. The purpose of the current study was to verify that the use of current wearable technologies for analysis of pulmonary oxygen uptake ($\dot{V}O_2$) dynamics during a pseudorandom ternary sequence (PRTS) over-ground walking protocol is a suitable procedure for the investigation of the aerobic response in more realistic settings. A wearable accelerometer located at the hip assessed the magnitude of the input changes delivered to the aerobic system. Eight adults (24 ± 4 yr old, 174 ± 7 cm, and 71.4 ± 7.4 kg) performed two identical PRTS over-ground walking protocols. In addition, they performed on the cycle ergometer two identical pseudorandom binary sequence (PRBS) protocols and one incremental protocol for maximal $\dot{V}O_2$ determination. In the frequency domain, mean normalized gain amplitude (MNG in %) quantified $\dot{V}O_2$ dynamics. The MNG during PRTS was correlated ($r = -0.80$, $P = 0.01$) with the $\dot{V}O_2$ time constant (τ) obtained during cycling. The MNG estimated during PRBS was similar to the MNG estimated during PRTS ($r = 0.80$, $P = 0.01$). The maximal $\dot{V}O_2$ correlated with the MNG obtained during the PRBS ($r = 0.79$, $P = 0.01$) and PRTS ($r = 0.78$, $P = 0.02$) protocols. In conclusion, PRTS over-ground walking protocol can be used to evaluate the aerobic system dynamics by the simultaneous measurement of $\dot{V}O_2$ and hip acceleration. In addition, the aerobic response dynamics from PRBS and PRTS were correlated to maximal $\dot{V}O_2$. This study has shown that wearable technologies in combination with assessment of MNG, a novel indicator of system dynamics, open new possibilities to monitor cardiorespiratory health under conditions that better simulate activities of daily living than cardiopulmonary exercise testing performed in a medical environment.

oxygen uptake kinetics; wearables; accelerometer; pseudorandom binary sequence; pseudorandom ternary sequence

THE EVALUATION OF THE AEROBIC RESPONSE to exercise stimulus provides valuable information regarding the aerobic system integrity (19, 41). Abnormal aerobic responses during exercise might occur before the clinical detection of noncommunicable diseases (12), arousing interest in the development of tools for the aerobic system dynamics assessment in real-life scenarios.

The mathematical characterization of the pulmonary oxygen uptake ($\dot{V}O_2$) in response to work rate changes can produce parameters in the time and frequency domains related to aerobic system dynamics (10, 19, 42). The time-domain analysis typically uses a single, or multiple, repetitions of step work rate change protocol, and the $\dot{V}O_2$ response is fitted with exponential functions. The time course of the $\dot{V}O_2$ response is reported by time constants. Time-domain approaches require specific laboratory conditions and classically deal with issues related to the reliability of the estimated parameters (19) due to low signal-to-noise ratio associated with $\dot{V}O_2$ measured at the mouth (24), as well as the high degree of freedom of the models chosen (4, 26).

A faster aerobic system dynamics, characterized by a faster $\dot{V}O_2$ adjustment, is classically associated with a better aerobic fitness (29, 35, 36) and disease prognosis (5, 33, 37). On the other hand, slower aerobic responses are related to a higher lactate production (23) and seem to impact functional mobility (1). As a marker for primary prevention of noncommunicable diseases (12), the early identification of the slowing of aerobic system adjustment has considerable potential in the future of health care.

The frequency-domain analysis of $\dot{V}O_2$ dynamics, while also constrained to date to laboratory conditions, has some advantages over the time-domain approach in that there are no assumptions regarding the model used to fit the data (10). Instead of a single process that requires the signal to fit into a predefined complex model, the frequency-domain method fits the data set into multiple sinusoidal functions. This process allows the investigation of the $\dot{V}O_2$ dynamics at different stimulus frequencies, thereby enabling a more detailed understanding of the aerobic system control during dynamic exercise transitions (15, 21). In addition, the system dynamics parameters obtained through frequency-domain analysis are less susceptible to nonperiodic signals, such as the intrabreath noise and can be used to fit signals obtained from random exercise stimuli. The normalized amplitudes of each frequency can be averaged together into a single parameter (named mean normalized gain, MNG) that describes the overall system dynamics and is conceptually similar to time constant (τ) from time-series analysis. As opposed to constant work rate protocols used for time-domain analysis, the optimal $\dot{V}O_2$ dynamics investigation by frequency domain requires more variation in the exercise protocols (21).

Cycle ergometer experiments allow the precise control of the work rate and, therefore, a better control over the external stimulus (defined as system input). However, cycling is not a

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common activity of daily living, and it is not widely used between different populations and cultures. On the other hand, walking is a universal physical activity that is performed in almost all age groups and health conditions. Thus, the aerobic response investigation during walking is more applicable for the general population. Although treadmill walking can be a sufficient method to study the physiology of walking with precise control of speed and grade, it is not a completely realistic approach to represent over-ground walking.

The purpose of this study was to analyze the aerobic system dynamics during pseudorandom ternary sequence (PRTS) over-ground walking protocol to better simulate activities of daily living compared with traditional exercise testing performed in a medical environment. An accelerometer (ACC) located at the hip was used to assess the magnitude of the input changes delivered to the aerobic system. The aerobic system dynamics obtained during the proposed over-ground walking protocol were compared with a controlled cycling protocol and related to maximal $\dot{V}O_2$. The hypothesis of this study was that the aerobic system dynamics determined from frequency-domain analysis during a novel over-ground walking protocol would correlate with kinetics parameters from cycle ergometry testing. The results of this study of $\dot{V}O_2$ response to changes in ACC could set the stage for future assessment of the $\dot{V}O_2$ dynamics based on random physical activities typically encountered in daily living.

METHODS

Study design. Eight healthy active nonathletic young men (24 ± 4 yr old, 174 ± 7 cm, and 71.4 ± 7.4 kg) participated in the study. This study was approved by the Office of Human Research of the University of Waterloo and was in agreement with the Declaration of Helsinki.

For the first visit, exercise was performed on the cycle ergometer (Lode Excalibur Sport, Lode B.V., Groningen, The Netherlands). After warm-up, participants completed two consecutive pseudorandom binary sequence (PRBS) protocols (10, 21, 44) followed by an incremental protocol (25 W/min). From the incremental protocol responses, the individual maximal $\dot{V}O_2$ (40.7 ± 6.6 $ml \cdot min^{-1} \cdot kg^{-1}$) and gas exchange threshold (GET; 28.2 ± 7.2 $ml \cdot min^{-1} \cdot kg^{-1}$) were identified. The moving seven-breath average $\dot{V}O_2$ and carbon dioxide output signals were used to identify the maximal $\dot{V}O_2$ and the GET. The maximal $\dot{V}O_2$ was considered as the average of the last 15 s before recovery following the criteria: *1*) respiratory exchange ratio higher than 1.1 and *2*) identification of $\dot{V}O_2$ plateau with an increase in work rate (18). The GET was obtained for each participant by a standard method (3). One participant was excluded from the GET analysis due to signal lost in the middle of the incremental test. The $\dot{V}O_2$ at GET corresponded to 70.7 ± 9.3% of the maximal $\dot{V}O_2$.

After a minimum of 1 wk, participants performed two PRTS over-ground walking protocols separated by a 30-min resting period. Before the first PRTS, participant's walking speed was determined from a timed 15-m segment, while they walked at three selected cadences: 75, 105, and 135 steps/min. This procedure was repeated three to four times for a better reliability.

Pseudorandom binary sequence protocol. A four-stage digital shift register was used to generate the pseudorandom binary sequence protocol (PRBS) protocol, as previously reported (21). The PRBS was composed of 15 units, each of 30 s of duration, totaling 450 s of protocol length. The work rate varied between 25 and 100 W. The cycling cadence was maintained between 60 and 65 rpm. An extra 200 s of the sequence was added at the onset of the PRBS protocol as a warm-up and was excluded a priori, and then two identical sequences were completed.

Pseudorandom ternary sequence protocol. The pseudorandom ternary sequence protocol (PRTS) protocol included a variation of three walking cadences (75, 105, or 135 steps/min). The choice to select a protocol with three levels of exercise was based on the frequent change in speed of walking during real-life scenarios. These walking cadences were chosen on the basis of the variation of approximately ±30% of the normal average cadence (39). The PRTS protocol generation was based on the approach proposed by Peterka (34). The number of the shift register units was set at 3 to obtain a 13-min protocol (or 780 s) with a unit length of 30 s (Fig. 1*A*), thereby allowing for two repetitions of the protocol in the same visit. An additional 300 s of PRTS was added to the beginning of the protocol as a warm-up and excluded a priori; therefore, each PRTS total length was 18 min (or 1,080 s). Because of the PRTS protocol length, a resting period of 20 min was performed between both PRTS protocols.

The shift register outputs (0, 1, or 2) were converted into target cadences (Fig. 1*A*). A PRTS metronome audio file was created in Audacity 2.0.5 software (Carnegie Mellon University, Pittsburgh, PA) and listened to through ear buds to set the walking pace. The actual walking cadence performed by the participant was verified by a post hoc examination of walking cadence.

Data acquisition. The $\dot{V}O_2$ was measured with a portable metabolic system (K4b^2, COSMED, Rome, Italy). The chemical galvanic O_2 sensor, infrared absorption CO_2 sensor, and the low-resistance turbine rotor flowmeter were calibrated following manufacturer's specifications before every data collection. When appropriate, the METS was calculated by $\dot{V}O_2 \cdot kg^{-1} \cdot 3.5^{-1}$.

The three-axis hip ACC data were obtained from a previously validated (40) smart shirt (Hexoskin, Carré Technologies, Montréal, Canada). The ACC sample rate (64 Hz) and resolution (0.004 g) were sufficient to capture all expected movements during the proposed walking protocol and activities of daily living (6).

Data analysis. The step cadence was converted into speed for each participant using the individual linear regression between the cadence and walking speed (38). The ACC raw data were converted to the total vector magnitude (9) by $ACC = \sqrt{x^2+y^2+z^2}$; where x, y, and z are the vertical, longitudinal, and lateral ACC axis, respectively.

The system inputs (walking speed and ACC for PRTS and watts for PRBS) and output ($\dot{V}O_2$) collected during each two PRBS and two PRTS were linearly interpolated on a second-by-second basis, time aligned and ensemble averaged to obtain a single response per participant for each protocol. The discrete Fourier transformation (DFT) was used to convert the finite time series response into frequency space. To adhere to the linearity principle (15), the highest frequency analyzed was restricted at the maximum of 8.88 and 8.97 mHz for the PRBS and PRTS protocol, respectively. This upper frequency bound was selected so that no frequency amplitude values from above 10 mHz were included in any of the interpolated estimates (described below). The DFT algorithm fitted the data into sinusoidal functions by calculating the sine and cosine coefficients by the following equation:

$$y_{(t)} = a_{DC} + 2 \cdot \sum_{h=1}^{7} [A_h \cdot \cos(2\pi \cdot h \cdot f_1 \cdot t) + B_h \cdot \sin(2\pi \cdot h \cdot f_1 \cdot t)],$$

where y is the time-series signal to be fitted, t is the time, a_{DC} is average response (i.e., system DC_{offset} or zero-frequency component), f_1 is the fundamental frequency (2.22 and 1.28 mHz for PRBS and PRTS, respectively), and A_h and B_h are the cosine and sine amplitude coefficients, respectively. The parameter h is the harmonic number (continuous and even integer numbers for PRBS and PRTS, respectively). From A_h and B_h, the total amplitude was calculated for each harmonic h by $Amp_h = \sqrt{A_h^2+B_h^2}$ (21). Because f_1 was different between protocols, the input and output *Amp* responses were linearly interpolated for each protocol at a common frequency range of 2.5 to

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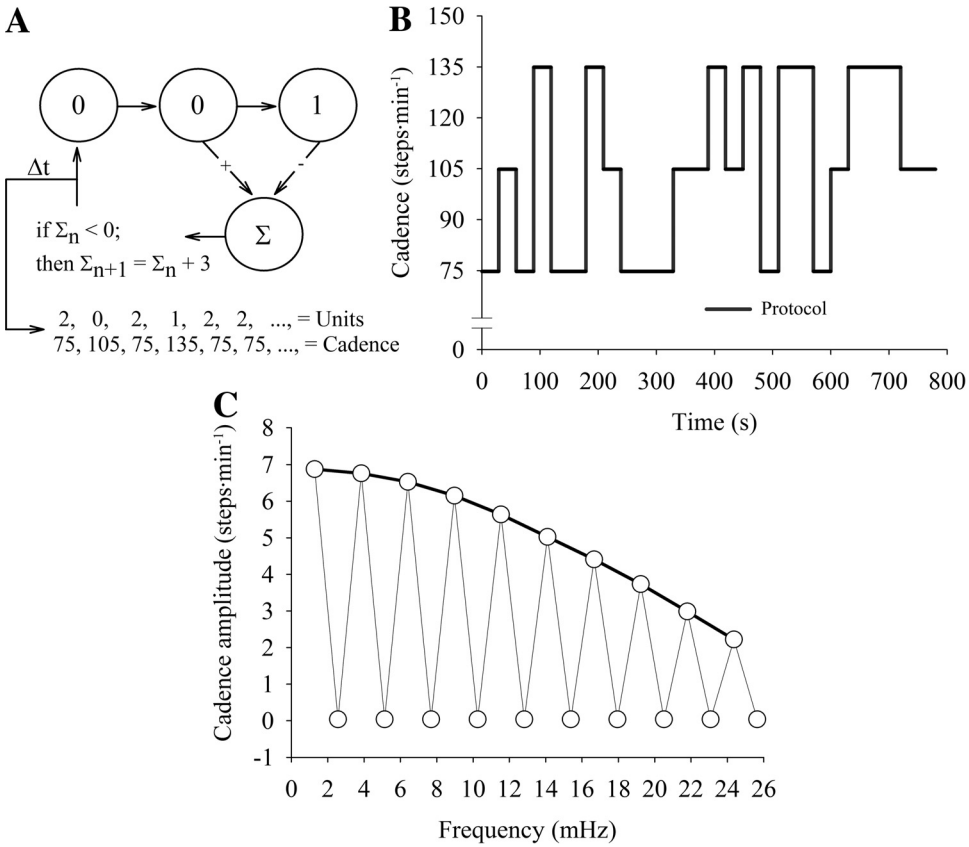

Fig. 1. *A*: selected structure of the shift register used to generate the pseudorandom ternary sequence protocol. The module addition feedback (Σ) sums the negative value of the third stage with the second stage and tests the "if" statement. This new signal is inserted into the first stage and shifts the entire system to the right. The unit value is held for "Δ*t*"s. The unit values (0, 1, or 2) were transformed in the selected walking cadences (0 = 105, 1 = 135, and 2 = 75). The six-unit values shown in *A* correspond to the first six stages in *B*. The protocol in the time domain (in *B*) was transformed into frequency domain by Fourier transformation, and the amplitude for each corresponding sinusoidal function was computed across the frequencies (as displayed in *C*). Also in *C*, as a characteristic of pseudorandom ternary sequence (PRTS) protocols, the stimulus energy decreases to zero at even harmonics (22).

8.5 mHz, with a resolution of 0.5 mHz. This resulted in 13 paired-between-protocol frequencies. To eliminate the influence of the intrasubject variability of the system static gain (15, 16), the system gain (Amp_{output}/Amp_{input}) was normalized as a percentage of the *Amp* gain at the lowest common frequency (i.e., 2.5 mHz) (15). Finally, the MNG that describes the overall temporal system dynamics was obtained for both protocols by the average of the interpolated normalized gains between the same common frequencies (2.5 to 8.5 mHz).

To test the ability of the novel indicator MNG to extract the same dynamic characteristics of the $\dot{V}O_2$ response as observed in time-domain analysis from the time constant (t), the $\dot{V}O_2$ ensemble-averaged data during the PRBS protocol were submitted to time-domain analysis. The timescale was shifted to align *time 0* with the onset of the second 100-W step at 120 s of the PRBS protocol. The data window was composed of 15 s of baseline (at 25 W) followed by 120 s of constant work rate at 100 W. This data set was the longest period without work rate variation within the PRBS protocol and, thus, the best data window for time-domain analysis. The first 20 s of data related to the cardiodynamic component (2) were excluded from the analysis. For one participant, only 91 s after baseline was considered due to an unexpected $\dot{V}O_2$ overshoot. The remaining data were fitted by a monoexponential model following a standard method (20, 42) to obtain τ and the steady-state $\dot{V}O_2$. As displayed in Fig. 2, the MNG during PRTS was negatively correlated with τ considering ACC ($r = -0.75$, $P = 0.03$, and $n = 8$) or walking speed ($r = -0.80$, $P = 0.01$, and $n = 8$) as system input. Thus, higher MNG values are associated with faster $\dot{V}O_2$ dynamics (i.e., lower τ). Data analysis was performed by a certified (no. 100-314-4110) LabView-associated developer (National Instruments, Austin, TX).

Statistical analysis. Friedman repeated-measures ANOVA was used to compare the interpolated normalized gain *Amp* between the different tested frequencies. Student-Newman-Keuls was used as a post hoc test. The MNG obtained from PRBS and PRTS protocols were compared by Student's *t*-test. Linear correlation was measured by Pearson product-moment correlation coefficient (r). Statistical significance was set at a level $\alpha = 0.05$. Statistical analysis was performed in SigmaPlot 12.5 software (Systat Software, San Jose, CA).

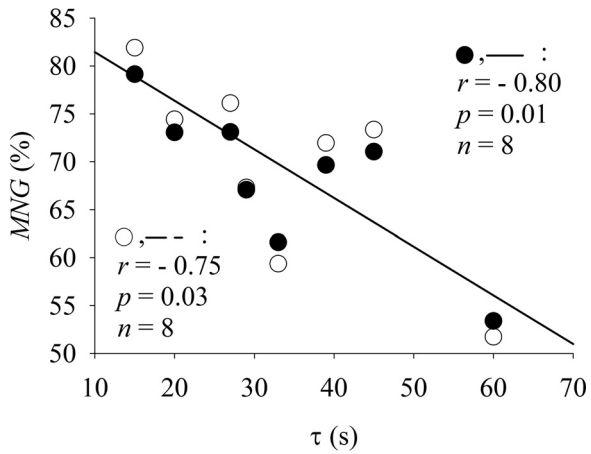

Fig. 2. Relationship between the oxygen uptake time constant τ obtained during cycling and the mean normalized gain amplitude (MNG) obtained during PRTS over-ground exercise protocol. The MNG was estimated on the basis of hip acceleration (○) or walking speed (●) as system inputs during PRTS. This relationship followed a linear pattern. r, Pearson's correlation coefficient; P, statistical significant level; and n, sample size.

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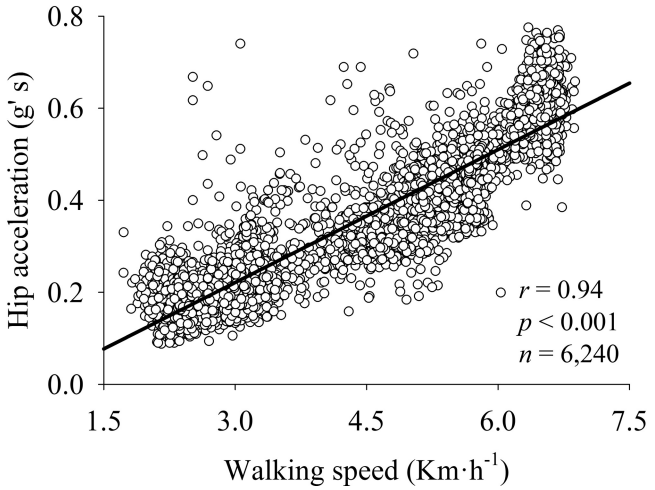

Fig. 3. A linear correlation between walking speed and hip acceleration is shown.

RESULTS

Figure 3 illustrates the linear relationship between walking speed and hip acceleration during the PRTS protocol (780 samples per participant, 6,240 in total). The hip acceleration presented a strong positive correlation ($r = 0.94$ and $P < 0.001$) with the walking speed. The ACC data can be used in place of directly measured walking speed.

Time series. The time-series mean $\dot{V}O_2$ response of all tested participants ($n = 8$) is displayed in Fig. 4, *A* and *B*, for the PRBS and PRTS protocols, respectively. The work rate (watts) was used as the system input during cycling (PRBS), and the hip acceleration was displayed as a system input during over-ground walking (PRTS). As described in Fig. 4*C*, the distribution of the metabolic equivalent varied between three and eight METS during both protocols, and this range was similar to the metabolic demand of moderate activities of daily living (14). The higher relative incidence of METS (≈30% of the samples) was ≈5.5 and ≈3.8 METS for PRBS and PRTS, respectively. The METS were consistently lower during PRTS in comparison to PRBS, indicating that the metabolic demand during PRTS was also restricted to moderate-intensity exercise. In addition, the average of the steady-state $\dot{V}O_2$ estimated from the monoexponential data modeling during PRBS (23.9 ± 2.9 $ml \cdot kg^{-1} \cdot min^{-1}$, 85.6 ± 10.4% of the $\dot{V}O_2$ at GET, and 59.8 ± 8.4% of the maximal $\dot{V}O_2$) indicated that the 100-W work rate correspond to moderate-intensity exercise. Therefore, considering similar dynamics between PRBS and PRTS (further demonstrated), the PRTS protocol also constrained to moderate exercise intensity.

Frequency domain. The system dynamics characterization by frequency-domain analysis was based on the study of the input-output relationship (gain) across different frequencies. Figure 5 shows the comparison of the interpolated normalized gain *Amp* obtained from PRBS and PRTS protocols across the range of 13 selected frequencies. The influence of the stimulus frequency over the aerobic system response was similar between protocols, indicating similar $\dot{V}O_2$ dynamics between these two exercise modalities (cycling vs. walking). As ex-

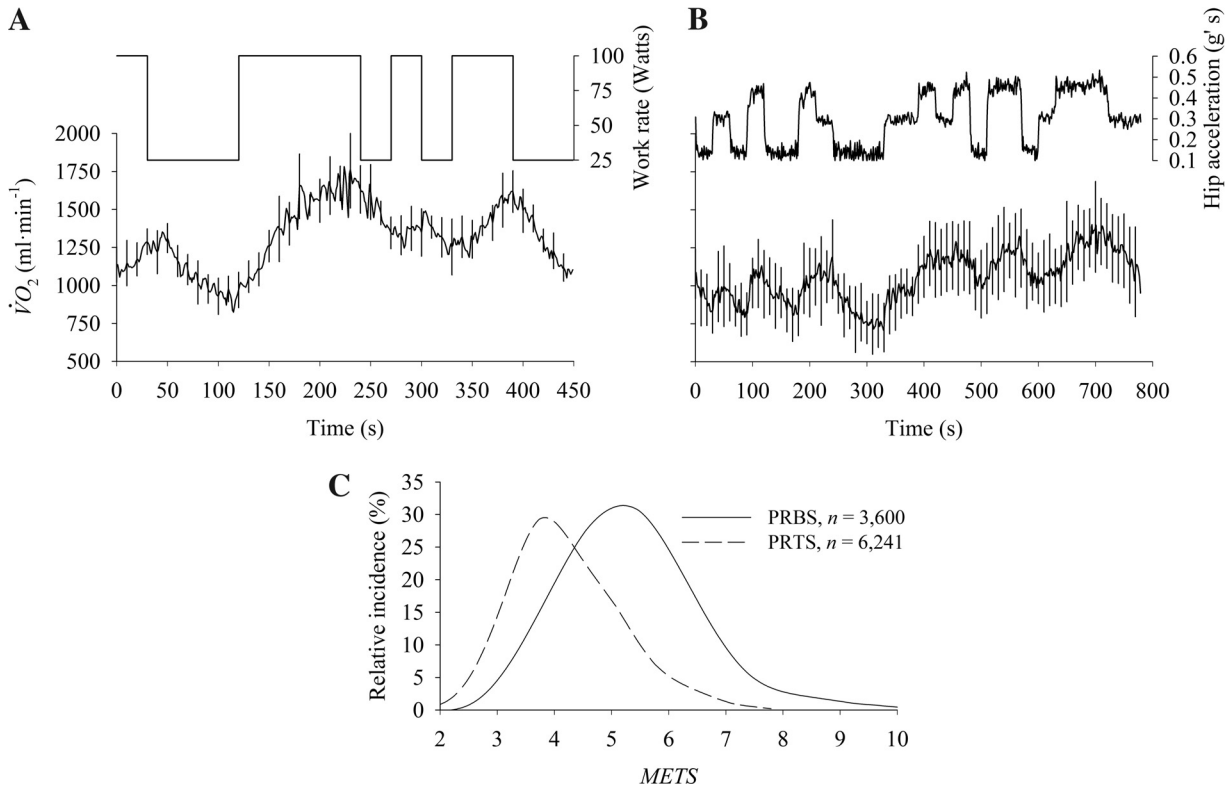

Fig. 4. Mean (—) ± SD (vertical bars at 10-s intervals) of the oxygen uptake response ($\dot{V}O_2$) of all participants ($n = 8$) during pseudorandom binary sequence (PRBS; *A*) cycling protocol and PRTS (*B*) over-ground walking protocol. The upper portion of each panel describes the work rate and hip acceleration of a representative participant during PRBS and PRTS protocols, respectively. *C*: distribution of the metabolic equivalents (METS) during PRBS and PRTS.

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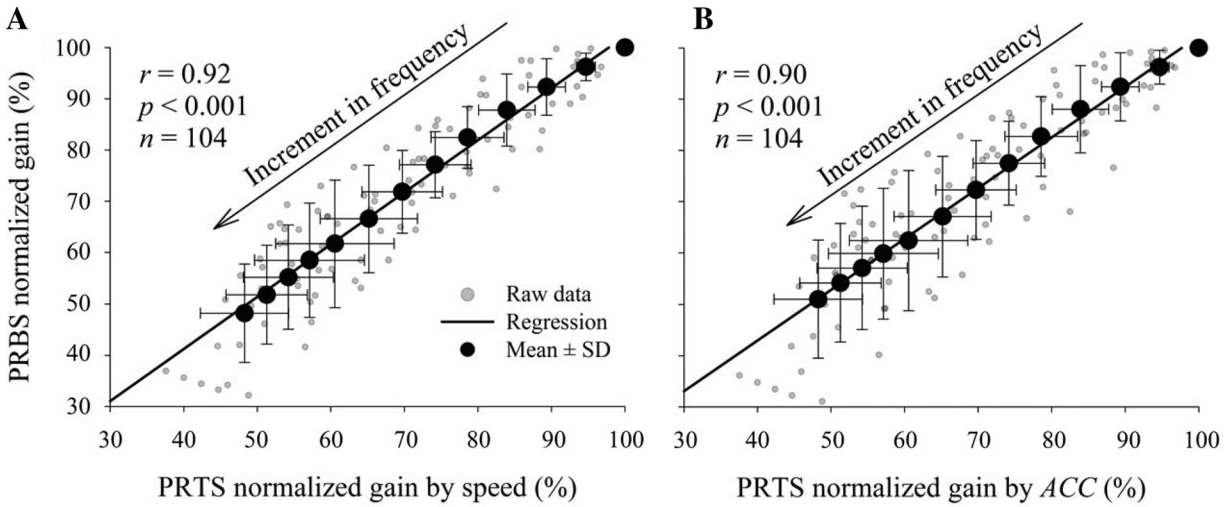

Fig. 5. Correlation of the effect of the input frequency increment (arrow) over the normalized system gain between PRBS and PRTS protocols. Graphs display the mean ± SD response of all participants (n = 8) at each frequency (n = 13) and the raw data for all participants and frequencies (n = 104). The Pearson correlation coefficient (r) and the statistical significance level (P) were calculated on the basis of the raw data from all participants and frequencies. The aerobic system gain during PRTS was estimated using walking speed (*A*) or hip accelerometer (ACC) (*B*) as system input.

pected, the system gains in both protocols statistically ($P <$ 0.001) decreased as the frequency increased. The strong linear correlation ($r > 0.9$ and $P < 0.001$) between PRBS and PRTS was not different whether the PRTS protocol used walking speed (Fig. 5*A*) or ACC (Fig. 5*B*) signal as system input.

Mean normalized gain. The mean normalized gain (MNG), as an index for the aerobic system dynamics evaluation, was estimated for each participant based on the average of the normalized gains. The MNG was statistically ($P > 0.05$) similar between PRBS and PRTS. During the PRTS protocol, the MNG estimated using walking speed as system input was strongly positively correlated ($r = 0.99$ and $P < 0.001$) with the MNG estimated from ACC (Fig. 6*A*). The MNG during PRTS, from either walking speed or ACC, was statistically similar and strongly positively correlated to the MNG obtained from the PRBS protocol (Fig. 6*B*).

Figure 7 displays the correlation between the MNG estimated from PRBS (Fig. 7*A*) and PRTS (Fig. 7*B*) protocols with the maximal $\dot{V}O_2$ obtained during incremental cycling protocol. The MNG during PRBS protocol was strongly positively correlated ($r = 0.79$ and $P = 0.01$) with maximal $\dot{V}O_2$. The MNG and maximal $\dot{V}O_2$ were strongly positively correlated during PRTS when walking speed ($r = 0.81$ and $P = 0.01$) or ACC ($r = 0.78$ and $P = 0.02$) signals were used as system inputs.

DISCUSSION

In alignment with our initial hypothesis, the aerobic system dynamics in response to a novel over-ground walking protocol correlated with the aerobic dynamics during cycle ergometry testing. The oxygen uptake kinetics assessed by the mean

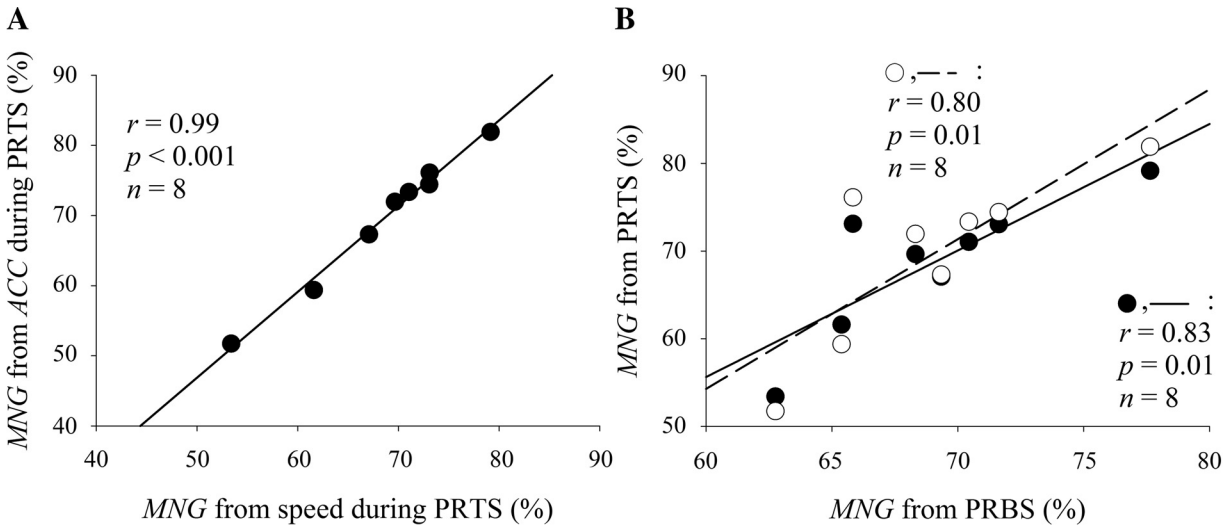

Fig. 6. MNG obtained from the oxygen uptake response during PRTS and PRBS protocols. *A*: correlation between the MNG estimated based on hip acceleration (ACC) and walking speed as system inputs during PRTS. *B*: correlation between MNG estimated during PRBS and PRTS protocols. The MNG was estimated using ACC (○) or walking speed (●) as system inputs during PRTS.

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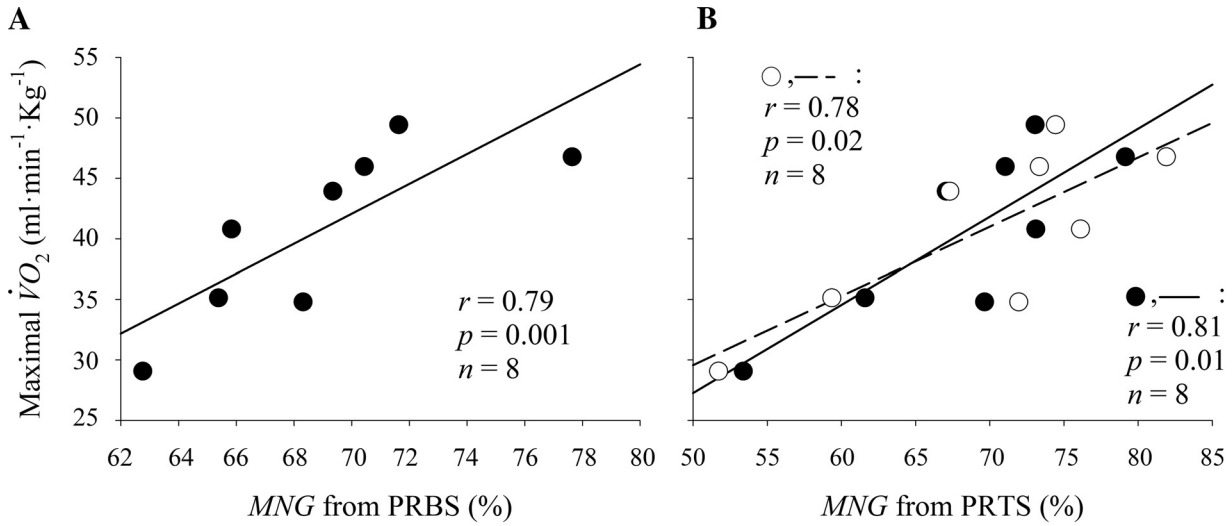

Fig. 7. *A*: correlation between maximal oxygen uptake ($\dot{V}O_2$) and MNG obtained from the oxygen uptake response during PRBS. *B*: correlation between maximal $\dot{V}O_2$ and MNG obtained from the oxygen uptake response during pseudorandom ternary sequence (PRTS). The MNG was estimated using ACC (○) or walking speed (●) as system inputs during PRTS.

normalized gain during the pseudorandom ternary sequence walking test correlated with the traditionally determined time constant measured during cycling exercise. The current study has added to the understanding of $\dot{V}O_2$ kinetics during dynamic transitions in exercise and provided a foundation upon which $\dot{V}O_2$ kinetics could be studied during variations in the speed of over-ground walking, such as might occur during normal activities of daily living. To our knowledge, this was the first time that a PRTS protocol was used to study $\dot{V}O_2$ kinetics. Our results showed that hip accelerometers can be used as a proxy of work rate during random activities and, therefore, as a system input for aerobic system dynamics investigation. In addition, the $\dot{V}O_2$ dynamics, assessed by the MNG, were similar between PRTS and PRBS, and correlated with maximal $\dot{V}O_2$.

The advancements of low-cost and comfortable wearable sensors allowed the acquisition of intensive longitudinal biological data during unsupervised activities of daily living (27, 30). Accelerometers are one of the most common wearable sensors used to infer the external work originating from spontaneous physical activity (43). Therefore, this sensor has the potential to be used as a system input for the study of the aerobic system dynamics during realistic activities, which might expand physical fitness evaluation far beyond the laboratory constraints. In this study, the hip accelerometer signals were successfully validated in comparison to walking speed data (Fig. 3) that is correlated to metabolic demand (17, 39). However, the detection algorithm used to estimate walking cadence and, therefore, the walking speed, may fail to identify steps during complex body movements, as expected during activities of daily living (25). Therefore, the metabolic cost and system input during activities of daily living might be better characterized by the hip accelerometer in comparison to walking speed or cadence, since it is not necessary to detect cyclic events (steps). In addition, the accelerometer was located closer to the body center of mass (i.e., hip) with a higher correlation to the metabolic demand during physical activity (8).

Walking is considered a common activity of daily living that requires a high O_2 demand (and, therefore, greatest increase in $\dot{V}O_2$) due to the high degree of muscle activation. This characteristic enables the clear differentiation of the $\dot{V}O_2$ response from the resting metabolic rate. During activities of daily living, the evaluation of the $\dot{V}O_2$ data during walking periods has the potential to be used to assess the aerobic system dynamics. To test this potential, the proposed PRTS protocol was used to simulate the step cadence changes that are expected to occur during realistic activities of daily living, although not with the same pattern. The $\dot{V}O_2$ dynamics in response to the PRTS protocol was similar to the well-established PRBS protocol (10, 21). However, the proposed PRTS protocol has more applicability for the general population since the work rate was delivered by over-ground walking.

Like PRBS, the $\dot{V}O_2$ adjustment during the PRTS protocol depends on the ability of the cardiorespiratory and muscular systems to provide and use O_2, respectively (19). This ability modulates the rate at which the $\dot{V}O_2$ increases after the exercise stimulus and faster dynamics responses are associated with better aerobic fitness (13, 36). Therefore, a better coupling between O_2 delivery and its utilization will directly influence the $\dot{V}O_2$ response during PRTS, allowing the identification of different aerobic responses in submaximal exercise (21, 44).

Despite the strong linear correlation between MNG and maximal $\dot{V}O_2$ ($r = 0.80$), the MNG was not able to account for all variation in maximal $\dot{V}O_2$ between participants. However, the maximal $\dot{V}O_2$ estimation, used as the gold standard method, also has an expected source of error (18). Nevertheless, a faster $\dot{V}O_2$ response (i.e., higher MNG) was observed in participants with higher maximal $\dot{V}O_2$, which indicates that MNG can be used, at least, as a complementary marker of the aerobic system integrity in association to maximal $\dot{V}O_2$.

Unlike a completely random protocol, PRTS protocols offer stimulus patterns optimized to study the physiological responses during exercise for a detectable and wide frequency range that is necessary for the precise $\dot{V}O_2$ kinetics analysis. The unit length for each PRTS stimulus was 30 s rather than 5 s

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(Fig. 1*B*). In addition, to be more reliable for walking and to maintain consistency between studies, the choice to use this work rate unit duration was based on previous findings (10, 21). These researchers (10, 21) studied the $\dot{V}O_2$ response during 30-s and 5-s unit PRBS protocols and found that the aerobic dynamics assessed through the $\dot{V}O_2$ data measured at the mouth level was less susceptible to hemodynamic distortions when a 30-s unit was used rather than a 5-s unit.

As demonstrated in Fig. 2, the MNG index has the characteristic of isolating temporal dynamics of a linear system from variable static gains between participants and exercise modalities. The strong correlation between the MNG obtained from the $\dot{V}O_2$ response to cycling PRBS and walking PRTS protocols (Fig. 6*B*) might suggest that the temporal characteristics of the aerobic system are controlled by the same fundamental mechanism(s) in these different exercises (7). Participants who presented a faster $\dot{V}O_2$ adjustment (i.e., higher MNG values) during cycling also presented faster dynamics during walking, even with the expected difference in muscle contraction regimen and fiber recruitment between these exercise modalities. However, the metabolic demand between both protocols was alike (Fig. 4*C*), differing only ~1.5 METS, indicating that further studies are necessary to evaluate how different absolute metabolic demands required by different exercise modalities might influence the $\dot{V}O_2$ dynamics in random physical activities.

Limitations. Some important limitations have to be considered from the evidence presented in this study. The range of maximal $\dot{V}O_2$ evaluated (29 to 49 $ml \cdot min^{-1} \cdot kg^{-1}$) was smaller in comparison to previous literature (10). Therefore, new studies are necessary to verify the association between the $\dot{V}O_2$ dynamics during submaximal dynamics exercise across a wider range of maximal aerobic power assessed by maximal $\dot{V}O_2$ (in athletes or disease state for example).

The frequency-domain $\dot{V}O_2$ kinetics analysis based on PRBS/PRTS stimuli might be "contaminated" by asymmetries between the temporal dynamics of the on- and off-transient phases (32). The MNG obtained from the Fourier transformation has the potential to identify different system dynamics based on random exercise protocols if the exercise is constrained to the moderate-intensity domain, where the $\dot{V}O_2$ on- and off-kinetics are symmetrical (31) and the dynamic system linearity is preserved for the evaluated frequencies (11, 15). Fortunately, the metabolic rate of the majority of activities of daily living fits into this range (14), so the current study was intentionally limited to light and moderate exercise (Fig. 3*C*).

Conclusion. Our data suggest that pseudorandom ternary sequence protocols can be used to evaluate the aerobic system dynamics. As an over-ground walking protocol, the proposed methodology is more applicable for testing the aerobic response in the general population. In addition, the aerobic response dynamics from PRBS and PRTS were correlated to maximal $\dot{V}O_2$ (Fig. 7), indicating a significant outcome of this study. Unlike cycling incremental protocols used to obtain maximal $\dot{V}O_2$, the proposed PRTS protocol is more functional, can be more broadly applied to normally sedentary or less healthy individuals, and reduces the risks associated with maximal exertion.

However, the MNG was obtained from an optimized exercise stimulus for frequency domain analysis, and further studies are needed to evaluate the consistency of the MNG to assess the aerobic system dynamics during completely random activities, such as unsupervised activities of daily living.

Perspectives and Significance

As a consequence of a slower energy supply by the aerobic system, a slower $\dot{V}O_2$ dynamics was previously associated with functional mobility impairments in older adults (1). The early detection of subclinical aerobic response depletion might be an indication of a decreased physiological reserve, which contributes to frailty (28). Therefore, indexes that describe "how fast" the energy demand is supplied by the aerobic system (such as the MNG) have the potential to be considered into models for the early detection of disease states. Additionally, wearable technologies (such as accelerometers and heart rate) are becoming more popular and less costly, allowing routine daily monitoring. The combination between wearables and new data processing techniques has direct applicability for disease prevention and for the evaluation of treatment progression.

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DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the authors.

AUTHOR CONTRIBUTIONS

T.B. and R.L.H. conceived and designed research; T.B. and R.L.H. performed experiments; T.B. and R.L.H. analyzed data; T.B. and R.L.H. interpreted results of experiments; T.B. and R.L.H. prepared figures; T.B. and R.L.H. drafted manuscript; T.B. and R.L.H. edited and revised manuscript; T.B. and R.L.H. approved final version of manuscript.

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